

Digital Twins for Predictive Maintenance in Automotive Aftersales

Maksim Bogachenkov

Syntegon

Stuttgart

wi23024@lehre.dhbw-stuttgart.de

Oliver Yanjie Feng

TRUMPF SE + Co. KG

Ditzingen

wi23030@lehre.dhbw-stuttgart.de

Joel Yeraí Martínez Campo

BITBW

Stuttgart

wi23194@lehre.dhbw-stuttgart.de

Abstract—This paper investigates the effect of digital-twin-based predictive maintenance on the automotive aftersales domain, focusing on the German passenger-car market. A model-based gap analysis is conducted to compare the interval-based current state against a digital-twin-enabled target state along the full customer journey. Based on this comparison, the business potential is quantified using a transparent assumption framework anchored in published German market data. The results show that proactive, data-driven service recommendations raise the repair capture rate from 40 to 70 percent, increasing wear-part gross profit by approximately 53 EUR per vehicle per year and 263,000 EUR annually across a 5,000-vehicle fleet. Customer touchpoints grow from 1.6 to roughly 2.7 per year, each representing an additional advisory and upselling opportunity. The analysis further identifies data access under the EU Data Act, rather than technology maturity, as the dominant adoption barrier. The findings establish a transparent, contestable model of the potential that dealerships and OEMs can use to evaluate and plan the integration of digital-twin-based predictive maintenance into their aftersales operations.

Index Terms—Digital Twin, Predictive Maintenance, Connected Vehicle, Machine Learning, Automotive Aftersales

I. INTRODUCTION

Today's vehicles are connected by default and continuously transmit operational data [1], [2], so the information required to assess their technical condition is largely available. Nevertheless, automotive aftersales remains reactive: even well-connected vehicles are serviced on fixed time or mileage intervals rather than on the actual condition of their components [3]. Wear parts—brake pads and discs, tyres, ignition components, or wipers—are typically replaced only after wear or failure, leading to unplanned repairs frequently carried out by independent third-party workshops [4]. For the dealership, which owns the customer relationship, this means losing repair revenue and valuable interactions, reducing opportunities for customer advice, retention, and upselling [5], [6].

Digital-twin-based predictive maintenance (PdM) can turn this data into foresight. A digital twin (DT) maintains a synchronized virtual representation of the vehicle [7], against which machine-learning (ML) models forecast the remaining useful life (RUL) of components and recommend replacement before failure—supporting closer customer retention at the dealership. While the building blocks of DT-based PdM are well studied in industrial settings [8], [9], their integration into automotive aftersales remains underexplored.

This paper therefore addresses the following research question: *what effect does the proactive use of digital twins and predictive maintenance have on the automotive aftersales domain?* Focusing on the German market and working with explicitly stated assumptions, the study (i) clarifies the interplay of DT and PdM in aftersales, (ii) compares the predictive target state with the interval-based current state along the full customer journey to identify where reactive servicing causes value leakage, and (iii) derives actionable recommendations for dealerships and original equipment manufacturers (OEMs).

II. RELATED WORK AND THEORETICAL BACKGROUND

A. Digital Twins

The concept of the digital twin was formalised by Grieves [7] in a white paper on manufacturing excellence, where it described a virtual replica of a physical product maintained in synchrony with its real-world counterpart throughout the entire asset lifecycle. Since then, the concept has expanded into aerospace, energy, and automotive domains [8], [10].

In the automotive domain, this general architecture maps directly onto the connected vehicle. The physical entity is the vehicle itself, equipped with on-board sensors and an On-Board Diagnostic (OBD) interface. Its virtual representation is a cloud-hosted model mirroring component states, degradation histories, and operating conditions. The synchronizing link is the over-the-air (OTA) telematics connection, which transmits telemetry data to the cloud and returns configuration updates to the vehicle [1], [2], [11]. Together, this setup transforms the vehicle from a periodically inspected asset into a continuously monitored system — a shift first documented for automotive applications by Zhang et al. [11].

B. Vehicle Maintenance Strategies

Vehicle maintenance strategies form a maturity hierarchy: reactive maintenance repairs only after failure [9]; preventive maintenance schedules service at fixed mileage or time intervals, leading to either premature replacement or undetected wear [9], [12]; condition-based maintenance (CBM) couples the service decision to sensor thresholds [12]; and PdM extends CBM by forecasting the future evolution of degradation, estimating the remaining time or distance until failure [9], [13].

At this scale, PdM is not a stand-alone technique but a service layer built on a DT of the vehicle. As described

by Lopes et al. [1] and El Hadraoui et al. [12], the DT provides the synchronized virtual representation against which sensor streams are interpreted, anomalies detected, and RUL estimates computed. In current automotive DT architectures, predictive analytics is therefore typically positioned as a dedicated microservice within the DT's analytics core [1], [14].

C. Machine Learning in PdM

Three main problem classes structure ML-based PdM. Classification assigns sensor patterns to discrete fault categories using Support Vector Machines, Random Forests, or Convolutional Neural Networks on time-frequency representations of vibration signals [9], [13]. Regression targets RUL estimation, dominated by Long Short-Term Memory (LSTM) networks for their ability to model long-range temporal dependencies in degradation signals [9], [14]. Anomaly detection operates without explicit labels, learning nominal behaviour and flagging deviations as failure precursors [9], [14]. AbouGrad et al. report classification accuracies of approximately 89% for vibration-based fault detection and 92% for LSTM-based RUL estimation in automotive contexts, though they note that these figures depend strongly on dataset quality and operating conditions [9]. In the aftersales context, RUL regression is the primary mechanism behind proactive service recommendations, whereas anomaly detection serves as an early-warning layer for components without established wear curves.

D. Automotive Aftersales

Aftersales encompasses post-purchase activities that enable customers to access supplier services and resolve product issues through continuous maintenance [15]. In the automotive context, it represents a dedicated business function comprising maintenance, repair, and warranty offerings to maximize product usage over the entire asset lifecycle while meeting economic and customer-related objectives [6], [16]. Building post-sales relationships strengthens customer loyalty, yields actionable insights, and creates upselling and cross-selling opportunities, making the aftersales market increasingly attractive as demand for higher-quality service expands [6].

Historically, once the warranty expires, customer engagement with the dealership tends to decline. Within the automotive industry, aftersales services represent a significant source of revenue, with profit margins substantially higher than those generated through vehicle sales [17]. From an economic perspective, the aftersales market has been identified as larger than the primary product sales market across various industries [10], [18], [19]. Retaining these customers therefore requires modern service processes [15].

Proactive aftersales improves customer satisfaction and retention by addressing issues before they are perceived by the customer, enabling more effective service interactions [5]. This requires tailoring services to customer needs through the integration of operational and informational data [15]. In the present context, DT-based PdM approaches support predictive interventions and improved asset lifecycle optimization, re-

sulting in reduced costs, increased customer satisfaction, and extended asset lifetimes [20].

E. Synthesis and Research Gap

The preceding subsections establish three building blocks: the DT as a synchronized vehicle model, PdM as a prognostic analytics layer built on top of it, and aftersales as the customer-facing business context in which service decisions are executed. Together, they form a potential end-to-end system in which the DT continuously provides component-state data to ML-based PdM models, whose forecasts are surfaced through aftersales processes to enable proactive service appointments and retain customers at the dealership.

While the referenced contributions establish the technical foundations of DT-based PdM, existing work concentrates either on algorithmic aspects in isolation [9], [13] or on industrial applications [8]. The integration of DT-based PdM into automotive aftersales business models, particularly regarding its implications for customer interaction, remains underexplored. This paper aims to address the research gap.

III. METHODOLOGY

Building on the theoretical background established in Section II, this paper applies a *model-based gap analysis* to answer the research question. A model-based rather than experimental approach is adopted for two reasons: deploying a prototype dealership system lies outside the scope of this paper, and no public in-service wear-part failure dataset for the German passenger-car market exists at the required resolution [9]. Explicit assumptions anchored to published German market statistics therefore substitute for empirical calibration, with the trade-off that the result is a transparent, contestable estimate of potential rather than a measured outcome.

The analysis proceeds in two stages. First, a lifecycle comparison (Section IV-A) contrasts the interval-based current state against a DT/PdM-enabled target state along the customer journey, identifying the structural points at which reactive servicing causes value leakage. Second, a quantitative potential assessment translates the identified gap into economic KPIs. A single economic lever—the repair capture rate c , defined as the share of addressable wear-part work retained at the franchised dealership—is used to quantify the gap; its target value is derived analytically from the mechanism assumptions rather than assumed directly, ensuring internal consistency. Finally, a one-way sensitivity analysis over c confirms that the economic case holds across a plausible parameter band [21].

IV. GAP ANALYSIS

A. Current- and Target-State Lifecycle Comparison

Figure 1 illustrates the interval-based current state along the customer journey from vehicle operation through service execution to post-repair follow-up, identifying the responsible actor at each stage and where reactive servicing causes value leakage. The processes are separated into a physical and a digital lane, making the extent of digital involvement directly comparable across the lifecycle. The corresponding DT/PdM-enabled target state is presented separately in Figure 2.

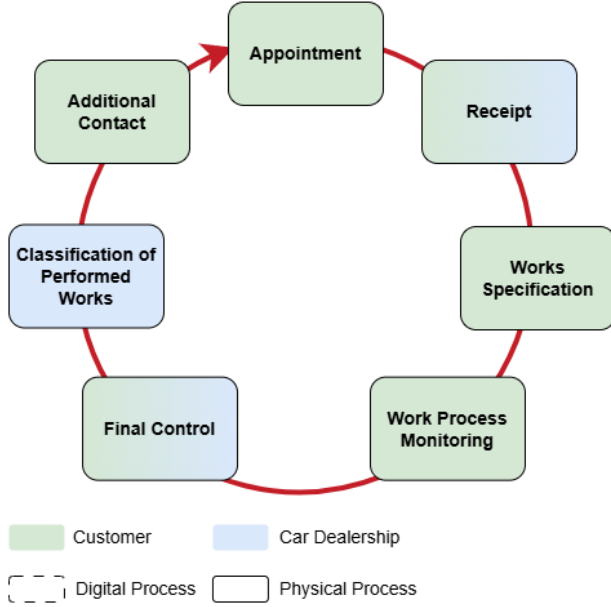


Fig. 1. Aftersales lifecycle: current interval-based state

a) Current state: In the interval-based current state, the customer journey is driven by fixed manufacturer schedules and visible failure signals. The vehicle operates without continuous component-state monitoring; no wear forecast is computed during normal driving. Consequently, most process steps are physical in nature and therefore cannot be represented in the digital lane. The lifecycle is initiated by the customer rather than the dealership: beginning with the appointment request, the subsequent process remains largely customer-driven. Service is triggered either by a time- or mileage-based reminder or by an acute fault, such as a warning lamp. Both represent late-stage maintenance triggers: maintenance intervals are set conservatively, leaving many components with substantial RUL, whereas fault-based triggers often occur only after wear limits have already been exceeded [9]. Most process steps are controlled by the customer and only sparsely supported by digital information because no data on the actual condition of vehicle components are available. As no proactive outreach precedes a fault, customers retain complete freedom to choose a repair provider at the moment of highest urgency, causing many unplanned repairs to be captured by independent third-party workshops [22]. This leakage intensifies with vehicle age—franchised-dealer share drops from 86 % for vehicles under three years old to just 39 % for those over six years [22]—and is further reinforced by cost-driven customer behavior, with 43 % of German vehicle owners performing only the minimum required maintenance [22]. The dealership's contact with the customer is therefore reduced to the statutory inspection cycle, limiting both retention and upselling opportunities.

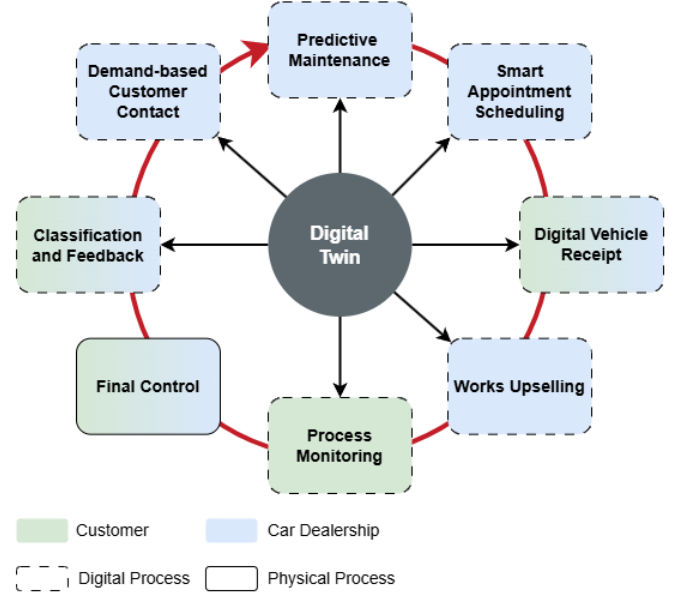


Fig. 2. Aftersales lifecycle: DT/PdM-enabled target state

b) Target state: The DT forms the core enabler of the digital process layer. It limits direct customer involvement to the necessary interactions, thereby improving the service experience compared to the current state. The lifecycle is initiated and steered by PdM forecasts rather than customer-triggered actions. Service requests, documentation, and process steps become digitally traceable, enabling additional data-driven analysis for upselling and service revenue optimization. Nevertheless, certain physical activities, such as vehicle inspection, remain outside the scope of full digitalization. In the DT/PdM-enabled target state, the vehicle's digital twin continuously aggregates sensor streams, OBD signals, operating parameters, and environmental conditions via OTA telemetry to maintain an up-to-date virtual representation of each monitored component [1]. ML models operating on this representation compute RUL for wear parts such as brake pads, discs, tyres, and ignition components [13], [14]. When the forecast horizon falls below a predefined threshold, the system generates a proactive service recommendation communicated via the OEM's connected-car interface or the dealership's CRM before any visible fault occurs [5], [20]. The appointment is scheduled at the customer's convenience, required parts are pre-ordered, and the workshop visit is transformed from an unplanned, fault-driven event into a planned, data-driven service interaction. After service completion, the digital twin resets the component baseline and resumes monitoring, thereby closing the lifecycle loop. The dealership thus captures wear-part replacements that would otherwise be lost to the independent aftermarket and gains structured touchpoints along the customer journey that are absent in the current state.

B. Assumption Framework

Because public, in-service failure datasets representative of this setting are largely absent [9], the comparison is built on an explicit assumption framework rather than on a proprietary dataset. Assumptions are divided into *context* assumptions, which fix the reference case (Table I), *wear-part* assumptions, which size the addressable work (Table II), and *mechanism* assumptions, which govern the transition from the current to the target state (Table III). Each is anchored to a published figure where one exists and flagged as a set value otherwise.

TABLE I
CONTEXT ASSUMPTIONS.

ID	Assumption	Value
A1	Reference vehicle	Mid-segment ICE passenger car, German market
A2	Annual mileage	13,140 km/yr [22]
A3	First-owner horizon	8 years [23]
A4	Workshop visits	1.6/yr [22]
A5	Considered wear parts	Brake pads/discs, tyres, wipers, 12 V battery, spark plugs
A6	Service gross margin	45 % (published range 40–55 %; set below the midpoint as a conservative estimate for mid-segment, non-premium dealers [23], [24])
A7	Franchised share	47 % [22]

The wear parts in A5 are deliberately limited to components whose wear is gradual, usage-driven, and not readily perceived by the driver—the class for which a forecast adds the most decision value. This class also dominates both between-interval failures and the most common defect categories in the mandatory German vehicle safety inspection (Hauptuntersuchung, HU: brakes, lighting, tires) [25]. Prescriptive maintenance and powertrain overhauls are out of scope. Typical replacement intervals and representative German market prices (parts plus labor, incl. VAT) are given in Table II for the reference vehicle (A1). All values are midpoints of published German market ranges [26]; actual intervals and prices vary substantially by vehicle class—compact and entry-level vehicles tend toward the lower end of each range, while premium and SUV segments tend toward the upper end. The sensitivity analysis brackets the residual uncertainty from this simplification.

TABLE II
WEAR-PART ASSUMPTIONS FOR REFERENCE VEHICLE A1 (MID-SEGMENT ICE). ALL VALUES ARE MIDPOINTS OF PUBLISHED GERMAN MARKET RANGES [26]; ACTUAL INTERVALS AND PRICES VARY BY VEHICLE CLASS

Part (A5)	Interval	Events/yr (at A2)	Price
Brake pads (per axle)	40,000 km	0.33	€ 250
Brake discs (+pads, front)	120,000 km	0.11	€ 450
Tyres (set of 4)	45,000 km	0.29	€ 600
Wiper blades (pair)	1 year	1.00	€ 25
12 V battery (fitted)	6 years	0.17	€ 150
Spark plugs (set)	60,000 km	0.22	€ 150

For the mechanism, a single lever is modeled: the *repair capture rate* c , the share of addressable wear-part work performed at the franchised dealership rather than lost to an independent workshop. Its complement $(1 - c)$ is the leakage. The current capture rate, forecast accuracy, and proactive-offer conversion are stated in Table III; the target capture rate is derived analytically instead of being assumed directly.

Of the leaked share $(1 - c_{\text{cur}})$, the fraction recaptured is the product of the forecast rate (a) and the proactive-offer conversion rate (k). The value $a=1$ is set because customers cannot distinguish a true-positive forecast from a false-positive recommendation. Every offer is therefore evaluated and accepted or declined at rate $a \cdot k$, regardless of whether the underlying wear has fully materialized:

$$\begin{aligned}
 c_{\text{tgt}} &= c_{\text{cur}} + (1 - c_{\text{cur}}) \cdot a \cdot k \\
 &= 0.40 + 0.60 \cdot 1.0 \cdot 0.50 \\
 &= 0.40 + 0.30 = 0.70.
 \end{aligned} \tag{1}$$

TABLE III
MECHANISM ASSUMPTIONS (CURRENT STATE → TARGET STATE).

ID	Assumption (Symbol)	Value	Basis
A8	Current capture rate (c_{cur})	0.40	<47 % share [22]
A9	PdM forecast accuracy (a)	1.0 (model)	reported 89–92 % [9]; set to 1 (see text)
A10	Proactive-offer conversion (k)	0.50	Conserv. est. [5], [27]
—	Target capture rate (c_{tgt} , deriv.)	0.70	$c_{\text{cur}} + (1 - c_{\text{cur}}) \cdot a \cdot k$, $a=1$

This construction keeps the estimate conservative and never approaches full capture. The conversion rate ($k = 0.50$, A10) reflects this: proactive, personalized maintenance contact from the customer’s own dealership is consistent with findings that proactive contact aids uptake and loyalty [5], [27]. Yet price competition from independent workshops and customer inertia toward switching service providers prevent full conversion, so the value of k is treated as a deliberate floor rather than a central estimate. The value of k is not empirically measured in this context; calibrating it against real customer-behavior data is identified as a direct future-work priority (Section VI-B).

C. Potential Assessment

The calculation proceeds in three steps: the addressable wear-part value per vehicle, the captured value under each state, and the resulting KPIs at vehicle, fleet and lifecycle level.

Step 1 — Addressable wear-part value: For each part i , the expected annual revenue is its replacement rate λ_i [yr^{-1}] multiplied by its unit price p_i ; both are from Table II, with λ_i computed on unrounded mileage (A2).

Summing across A5 yields the annual addressable wear-part revenue per vehicle:

$$\begin{aligned} V_{\text{wear}} &= \sum_i \lambda_i \cdot p_i \\ &= 82.1 + 49.3 + 175.2 + 25.0 + 25.0 + 32.9 \\ &\approx \text{€ } 389 \text{ per vehicle per year.} \end{aligned}$$

Step 2 — Captured value: Applying the capture rates from Table III:

$$\begin{aligned} \text{Captured}_{\text{cur}} &= c_{\text{cur}} \cdot V_{\text{wear}} = 0.40 \cdot 389 \approx \text{€ } 156 \text{ /veh/yr,} \\ \text{Captured}_{\text{tgt}} &= c_{\text{tgt}} \cdot V_{\text{wear}} = 0.70 \cdot 389 \approx \text{€ } 273 \text{ /veh/yr.} \end{aligned}$$

Applying the service gross margin (A6) gives the captured gross profit.

Step 3 — KPIs: Table IV summarises the comparison. Revenue and gross-profit figures are per vehicle per year; the fleet column scales the gross-profit delta linearly to a representative franchised group of 5,000 active customer vehicles.

TABLE IV
CURRENT-STATE / TARGET-STATE KPI COMPARISON. DELTA FIGURES COMPUTED DIRECTLY ON UNROUNDED INTERMEDIATE VALUES; CURRENT AND TARGET CELLS ROUNDED INDEPENDENTLY.

KPI	Current	Target	Δ
Repair capture rate	40 %	70 %	+30 pp
Wear-part revenue (/veh/yr)	€ 156	€ 273	+€ 117
Gross profit (/veh/yr)	€ 70	€ 123	+€ 53
Fleet GP, 5,000 veh (/yr)	€ 351k	€ 613k	+€ 263k
Customer touchpoints (/yr)	1.6	~2.7	+1.1
First-pass inspection rate	baseline	improved	qual.
Lifecycle retention [23]	~25 %	sustained	qual.

The proactive touchpoint count rises because each high-value forecast event (the wear parts in Table II excluding the wiper case, ≈ 1.1 events/yr) triggers an outreach contact that did not exist in the interval-based current state. This increases the meaningful contact frequency from 1.6 to about 2.7 per year, assuming forecast contacts fall outside scheduled service visits; overlap would reduce the net gain modestly. Each additional contact is also an advisory and upselling opportunity. The targeted parts—brakes and tires above all—are among the most common defect categories in the HU [25], so proactive replacement directly improves the customer’s first-pass inspection rate. This is a tangible and communicable benefit: 60 % of franchised-dealer customers currently report no additional service being offered at their last visit [22].

Sensitivity: The result depends most on the target capture rate. Varying c_{tgt} across a plausible band while holding V_{wear} and the 45 % margin (A6) fixed gives a per-vehicle gross-profit delta of:

$$\begin{aligned} c_{\text{tgt}} = 0.50 &\rightarrow \Delta \text{ gross profit} \approx \text{€ } 18 \text{ /veh/yr,} \\ c_{\text{tgt}} = 0.60 &\rightarrow \Delta \text{ gross profit} \approx \text{€ } 35 \text{ /veh/yr,} \\ c_{\text{tgt}} = 0.70 &\rightarrow \Delta \text{ gross profit} \approx \text{€ } 53 \text{ /veh/yr,} \\ c_{\text{tgt}} = 0.80 &\rightarrow \Delta \text{ gross profit} \approx \text{€ } 70 \text{ /veh/yr.} \end{aligned}$$

Even at the conservative end, the dealership captures additional gross profit of tens of euros per vehicle per year, scaling to a six-figure annual sum across a mid-size fleet. Moreover, compounding effects are likely given that 90 % of German vehicle owners consistently return to the same workshop [22]. The economic case is therefore robust to the assumptions rather than an artifact of any single optimistic value.

D. Limitations

The analysis is intentionally assumption-driven and carries corresponding limitations. First, the figures are illustrative: V_{wear} , the capture rates and the conversion factor are anchored to published market and accuracy data but are not fitted to an operational dataset, and no prototype was built; the contribution is a transparent model of the potential, not a measured outcome. The first-owner horizon (A3: 8 years) is an upper-bound lifecycle horizon; the DAT Report 2026 records a median holding period of 6.2 years [22], so fleet-lifetime totals are correspondingly optimistic. Second, the reported PdM forecast accuracy (≈ 90 %, the empirical basis of A9) is stated for transparency but does not enter the capture-rate formula. Because a proactive service recommendation looks identical to the customer whether the underlying forecast is correct or slightly premature, both true-positive and false-positive predictions generate the same type of offer, accepted or declined at rate k . Accuracy will improve continuously as in-service data accumulates [9], but the model outcome does not depend on reaching any particular accuracy threshold. A related long-run risk is trust erosion: if the system persistently recommends replacements the customer later judges premature, the effective conversion rate k will fall below the modelled floor over repeated interactions; the sensitivity analysis brackets this effect indirectly through the lower- c_{tgt} scenarios. Third, and most consequential for practice, the target state presumes the dealership can actually obtain the vehicle’s telemetry. Under the EU Data Act (applicable from 12 September 2025) the user is the vehicle owner or driver, who may authorise a third-party such as the dealership to receive in-vehicle data on fair, reasonable and non-discriminatory terms, while the OEM remains the data holder; where the data is personal, the GDPR additionally requires a lawful basis [28]. Access is thus contingent on customer consent, OEM provisioning in a usable format, and business-to-business (B2B) compensation — a non-technical adoption barrier that the model does not price in. Finally, two scope caveats apply: tyres, though included as a wear part, are often served by specialist channels and may leak independently of the dealership’s PdM capability; the shift to BEVs will erode ICE-specific wear-part volume and shift the addressable basket toward brakes, tyres, and the high-voltage battery.

V. DISCUSSION AND RECOMMENDATIONS

A. Interpretation of Results

The per-vehicle gross-profit uplift of € 53 per year scales to € 263k annually across a 5,000-vehicle fleet and compounds to approximately € 2.1 million over the eight-year first-owner

horizon (A3). The parts margin is, however, only the visible component of the value: each of the additional 1.1 contact events per vehicle per year is simultaneously an advisory opportunity and a loyalty-reinforcing interaction, consistently linked to increased repurchase intent at the originating dealership [5], [27]. The sensitivity analysis confirms robustness: even at $c_{\text{tgt}} = 0.50$, the fleet-level gain remains approximately €90k per year—a positive business case under any plausible conversion scenario.

The model deliberately quantifies only the benefit side: investment costs for the DT platform, CRM integration, and consent infrastructure are OEM- and dealer-specific and not publicly available. The gross-profit delta nevertheless defines a break-even threshold—at €53 per vehicle per year, the system pays off as long as its total running cost remains below roughly €4.40 per connected vehicle per month—giving practitioners a concrete benchmark against vendor pricing rather than a speculative return-on-investment figure.

B. The Non-Technical Barrier

The DT/ML/OTA technology stack is commercially mature and not the binding constraint [1], [9]. The real bottleneck is telemetry access: as outlined in Section IV-D, in-vehicle data reaches the dealer only with the owner’s consent and on OEM-provisioned terms [28], making the dealer dependent on two actors it does not control. The practical recommendation is therefore to build consent infrastructure—opt-in flows integrated into the CRM, clear customer value propositions—as a prerequisite rather than an afterthought, independently of any specific DT deployment decision.

C. Implications for Practice and Research

Practitioners should position DT-based PdM as a CRM extension rather than an IT project. The measurable value driver is the quality of proactive outreach, not forecast accuracy [5], [20]; appointment-scheduling integration and communication design should therefore precede algorithm tuning. A phased entry via OEM brands that already expose connected-car telemetry APIs allows conversion rates to be measured before a broader infrastructure commitment is made.

For researchers, the priority is empirical validation of the conversion rate k , which is modelled here but not fitted to operational data. A field study measuring opt-in and outreach acceptance rates would directly calibrate this parameter. A second priority is extending the framework to battery electric vehicles, where the high-voltage battery’s state of health replaces spark plugs as the dominant prognostic target.

Beyond the dealership’s business case, DT-based PdM exemplifies the shift from merely smart to wise information systems: replacing parts on evidence rather than on fixed intervals avoids premature replacement and the associated material waste, the explicit assumption framework keeps the decision model transparent and contestable, and consent-based data access keeps the customer in control of the decision loop.

VI. CONCLUSION & FUTURE WORK

A. Conclusion

This paper examined the effect of digital-twin-based predictive maintenance on automotive aftersales, focusing on the German passenger-car market. A lifecycle comparison showed that the transition from interval-based servicing to a DT/PdM-enabled target state redirects wear-part work from the independent aftermarket back to the franchised dealership by replacing reactive, fault-driven appointments with proactive, evidence-based service recommendations. A quantitative potential assessment, anchored to published German market data, translated this shift into concrete KPIs: under a conservative conversion assumption ($k = 0.50$), the repair capture rate rises from 40 % to 70 %, increasing gross profit by approximately €53 per vehicle per year and €263k annually across a 5,000-vehicle fleet. Customer touchpoints grow from 1.6 to roughly 2.7 per year, each an additional advisory and upselling opportunity. The sensitivity analysis confirmed the economic case is robust across the full plausible parameter range. The dominant adoption barrier is non-technical: access to in-vehicle telemetry requires customer consent and OEM provisioning under the EU Data Act and GDPR, a hurdle any real-world deployment must resolve. In answer to the research question: the effect of DT-based PdM on automotive aftersales is a measurable shift of wear-part revenue from the independent aftermarket to the franchised dealership, accompanied by a structural increase in customer contact frequency that amplifies retention and upselling value beyond the parts margin alone.

B. Future Work

Three directions stand out for future research. First, the assumption framework should be validated against an operational DT deployment, replacing modelled KPIs with measured outcomes and calibrating the conversion rate k under real customer-behaviour conditions. Second, the data-access challenge warrants dedicated study: viable consent architectures and OEM-dealer data-sharing agreements under the EU Data Act need to be developed and their effect on customer opt-in quantified. Third, the framework should be extended to battery electric vehicles, where the high-voltage battery’s state of health becomes the dominant prognostic target and the addressable wear-part basket shifts away from ICE-specific components such as spark plugs.

REFERENCES

- [1] J. Lopes, A. Reis, L. Barbosa, and J. Barroso, “A vehicle digital twin in the context of a software ecosystem,” in *2025 IEEE Vehicle Power and Propulsion Conference (VPPC)*, 2025, pp. 1–6.
- [2] R. Coppola and M. Morisio, “Connected car: Technologies, issues, future trends,” *ACM Comput. Surv.*, vol. 49, no. 3, Oct. 2016. [Online]. Available: <https://doi.org/10.1145/2971482>
- [3] V. Biletskyi, O. Ivanushko, A. Loboda, O. Buhaichuk, and R. Lovha, “Overview of information and communication technologies and vehicle monitoring systems and analysis of their possibilities for the formation and implementation of innovative technologies for maintenance and repair of vehicles,” *Avtozhliakhovik Ukrainy*, vol. 1, no. 273, pp. 8–16, 2023. [Online]. Available: http://journal.insat.org.ua/?page_id=4937&lang=en_GB

- [4] W. Alexander, J. Dempsey, S. Dayal, and J. Ark, "The secret life of factory service centers," *McKinsey Quarterly*, p. 106, 2002.
- [5] G. Challagalla, R. Venkatesh, and A. K. Kohli, "Proactive postsales service: When and why does it pay off?" *Journal of Marketing*, vol. 73, no. 2, pp. 70–87, 2009. [Online]. Available: <https://doi.org/10.1509/jmkg.73.2.70>
- [6] O. Sabbagha, M. N. A. Rahman, W. R. Ismail, and W. M. H. W. Hussain, "Impact of quality management systems and after-sales key performance indicators on automotive industry: A literature review," *Procedia - Social and Behavioral Sciences*, vol. 224, pp. 68–75, 2016. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S1877042816304852>
- [7] M. Grieves, "Digital twin: Manufacturing excellence through virtual factory replication," *White Paper*, 2014.
- [8] C. Wang, I.-S. Fan, and S. King, "A review of digital twin for vehicle predictive maintenance system," in *Proceedings of the SAE AeroTech Conference*, 2023, SAE Technical Paper 2023-01-1024.
- [9] H. AbouGrad, V. Batthula, and F. Riaz, "XAI digital twin predictive maintenance in the automotive industry: Challenges and opportunities," in *Proceedings of the 2026 IEEE 5th International Maghreb Meeting of the Conference on Sciences and Techniques of Automatic Control and Computer Engineering (MI-STA)*, Sebha, Libya, 2026.
- [10] R. Wise and P. Baumgartner, "Go downstream: The new profit imperative in manufacturing," *Harvard Business Review*, vol. 77, no. 5, pp. 133–141, 1999.
- [11] Y. Zhang, G. W. Gannt, M. J. Rychlinski, R. M. Edwards, J. J. Correia, and C. E. Wolf, "Connected vehicle diagnostics and prognostics, concept, and initial practice," *IEEE Transactions on Reliability*, vol. 58, no. 2, pp. 286–294, 2009.
- [12] H. El Hadraoui, N. Ouahabi, N. El Bazi, O. Laayati, M. Zegrari, and A. Chebak, "Toward an intelligent diagnosis and prognostic health management system for autonomous electric vehicle powertrains: A novel distributed intelligent digital twin-based architecture," *IEEE Access*, vol. 12, pp. 110 729–110 761, 2024.
- [13] Vinay Kumar R, Pavan Kumar K. K., D. Ramasamy, M. M. Noor, W. S. W. Harun, L. Samylingam, N. Aslfattahi, M. S. Kiai, and C. K. Kok, "Digital twin framework for predictive maintenance of autonomous vehicle suspension system," in *2025 IEEE 4th International Conference on Smart Technologies for Power, Energy and Control (STPEC)*, 2025, pp. 1–6.
- [14] Karthick G, S. Devakatte, U. Kumar, D. Muthukumar, J. M. A. Sebastian, and J. P. Wagh, "Simulating the future: AI-powered digital twins for intelligent eMobility and software-defined vehicles," in *2025 IEEE International Transportation Electrification Conference (ITEC-India)*, 2025, pp. 1–4.
- [15] S. Verstrepen, D. Deschoolmeester, and R. J. Berg, "Servitization in the automotive sector: Creating value and competitive advantage through service after sales," in *Global Production Management*, K. Mertins, O. Krause, and B. Schallrock, Eds. Boston, MA: Springer US, 1999, pp. 538–545. [Online]. Available: http://link.springer.com/10.1007/978-0-387-35569-6_66
- [16] J. R. Balinado, Y. T. Prasetyo, M. N. Young, S. F. Persada, B. A. Miraja, and A. A. N. Perwira Redi, "The effect of service quality on customer satisfaction in an automotive after-sales service," *Journal of Open Innovation: Technology, Market, and Complexity*, vol. 7, no. 2, p. 116, 2021. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S219985312200885X>
- [17] N. Saccani, L. Songini, and P. Gaiardelli, "The role and performance measurement of after-sales in the durable consumer goods industries: an empirical study," *International Journal of Productivity and Performance Management*, vol. 55, no. 3, pp. 259–283, 2006. [Online]. Available: <http://www.emerald.com/ijppm/article/55/3-4/259-283/165392>
- [18] R. Bundschuh and T. Dezvane, "How to make after-sales service pay off," *McKinsey Quarterly*, vol. 4, pp. 116–128, 2003.
- [19] A. Fraß, "Impact of after-sales performances of German automobile manufacturers in China in service satisfaction and loyalty," in *1st International Conference on Business Management – New Challenges in Business Research – Conference Proceedings*. Editorial Universitat Politècnica de València, 2015, pp. 200–206. [Online]. Available: <http://ocs.editorial.upv.es/index.php/ICBM/11CBM/paper/view/1218>
- [20] V. González-Prida, C. P. Márquez, P. V. Gunckel, F. K. Rodríguez, and A. C. Márquez, "Digital transformation in aftersales and warranty management: A review of advanced technologies in I4.0," *Algorithms*, vol. 18, no. 4, p. 231, 2025. [Online]. Available: <https://www.mdpi.com/1999-4893/18/4/231>
- [21] D. F. Channon and T. Sammut-Bonnici, "Gap analysis," in *Wiley Encyclopedia of Management*, 1st ed., C. L. Cooper, Ed. Wiley, 2015, pp. 1–2. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/9781118785317.wcom120109>
- [22] Deutsche Automobil Treuhand GmbH, "DAT Report 2026," DAT, Tech. Rep., Jan. 2026, Analyse des Autojahres 2025. [Online]. Available: <https://www.dat.de/report/>
- [23] I. Maurer, N. Kramer, E. Scarinci, and M. Webster, "Optimizing dealer profitability with a service center tune-up," McKinsey & Company, Automotive & Assembly Insights, Nov. 2025, Published Nov. 19, 2025, accessed June 3, 2026. [Online]. Available: <https://www.mckinsey.com/industries/automotive-and-assembly/our-insights/optimizing-dealer-profitability-with-a-service-center-tune-up>
- [24] Zentralverband Deutsches Kraftfahrzeuggewerbe e.V., "Zahlen & Fakten 2024," ZDK, Tech. Rep., 2025, Ausgabe 2025; Service-Umsatz im Kfz-Gewerbe 36,3 Mrd. EUR. [Online]. Available: <https://www.kfzgewerbe.de/presse/publikationen/zahlen-fakten>
- [25] TÜV-Verband e.V., "TÜV-Report 2026," TÜV-Verband e.V., Tech. Rep., Nov. 2025, Basierend auf dem TÜV-Report 2026, abgerufen am 03.06.2026. [Online]. Available: <https://www.tuvsud.com/de-de/publikationen/tuev-report/die-haeufigsten-maengel>
- [26] T. Kroher, "Wartung, Service, Inspektion: Das müssen Sie wissen," ADAC e.V., Jul. 2025. [Online]. Available: <https://www.adac.de/rund-ums-fahrzeug/reparatur-pflege-wartung/wartung-inspektion/wartungsintervalle/>
- [27] A. G. González, "Service quality and repurchase behaviour in the Spanish automotive after sales business," *Journal of Relationship Marketing*, vol. 14, no. 3, pp. 239–267, 2015. [Online]. Available: <http://www.tandfonline.com/doi/full/10.1080/15332667.2015.1069456>
- [28] European Parliament and Council of the European Union, "Regulation (EU) 2023/2854 on harmonised rules on fair access to and use of data (Data Act)," Official Journal of the European Union, Dec. 2023, OJ L, 22 December 2023, accessed June 3, 2026. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2023/2854/oj/eng>

VERWENDUNG GENERATIVER KI-SYSTEME

Bei der Erstellung der eingereichten Arbeit haben wir die nachfolgend aufgeführten auf künstlicher Intelligenz (KI) basierten Systeme benutzt:

1) Claude (Anthropic)

Wir erklären, dass wir

- uns aktiv über die Leistungsfähigkeit und Beschränkungen der oben genannten KI-Systeme informiert haben,
- die aus den oben angegebenen KI-Systemen direkt oder sinngemäß übernommenen Passagen gekennzeichnet haben,
- überprüft haben, dass die mithilfe der oben genannten KI-Systeme generierten und von uns übernommenen Inhalte faktisch richtig sind,
- uns bewusst sind, dass wir als Autorinnen bzw. Autoren dieser Arbeit die Verantwortung für die in ihr gemachten Angaben und Aussagen tragen.

Die oben genannten KI-Systeme haben wir wie im Folgenden dargestellt eingesetzt:

Tabelle I. Einsatz von KI-Systemen

Arbeitsschritt	Eingesetzte(s) KI-System(e)	Beschreibung der Verwendungsweise
Formatierung und Satz	Claude (Anthropic)	Umwandlung von Fließtext in $\text{\LaTeX}$ (Tabellen, Formeln, Literaturverzeichnis)
Verifikation der Berechnungen	Claude (Anthropic)	Prüfung der Zwischenrechnungen (V_{wear} , Capture-Raten) auf Konsistenz
Sprachliche Überarbeitung	Claude (Anthropic)	Korrektur von Grammatik und Stil einzelner Abschnitte